

Equation Sheet

- Real GDP = $Y = C + I + G + NX$
- Nominal GDP = $P \times Y$
- GDP deflator = P
- Exchange rates: $e^{nom} = \frac{\text{Foreign Currency}}{\text{US Dollar}}$; $e^{pp} = \frac{\text{Price in country i}}{\text{Price in US}}$
- $GDP^{PPP} = \frac{\text{GDP in local currency}}{e^{pp}}$
- Real interest rate (approximation): $r = i - \pi$
- Inflation: $\pi_{t+1} = \frac{P_{t+1} - P_t}{P_t}$
- Expected real interest rate: $E[r] = i - \pi^e$
- Elasticity of y with respect to x : $\epsilon_{y,x} = \frac{\partial y}{\partial x} \frac{x}{y} = \frac{\partial \ln y}{\partial \ln x}$
- Growth rate of a variable y_t : $g_t = \frac{y_t - y_{t-1}}{y_{t-1}}$
- Production Function: $Y = AF(K, N)$ (A =TFP; K =Capital; N =Labor input)
- Labor's share of income = $\frac{wN}{Y}$
- Growth Accounting: $\frac{\Delta A}{A} = \frac{\Delta Y}{Y} - \epsilon_{Y,K} \frac{\Delta K}{K} - \epsilon_{Y,N} \frac{\Delta N}{N}$, where (for example) $\epsilon_{Y,K}$ denotes the elasticity of Y with respect to K .
- Marginal products of labor and capital:

$$MPN = \frac{\partial Y}{\partial N} = A \frac{\partial F(K, N)}{\partial N} = AF_N$$

$$MPK = \frac{\partial Y}{\partial K} = A \frac{\partial F(K, N)}{\partial K} = AF_K$$

- Cobb-Douglas production function: $Y = AK^\alpha N^{1-\alpha}$
- Capital accumulation/evolution equation: $K_{t+1} = (1 - \delta)K_t + I_t$
- Labor-consumption optimality – the “static first order condition”: $-U_N = \frac{1-\tau_N}{1+\tau_C} w U_C$
- Unemployment rate: $u = \frac{d}{d+f}$ (d =job destruction/separation prob.; f =job finding prob.)
- Unemployment rate: $u = \frac{U}{U+E}$
- Unemployment dynamics: $U_{t+1} = (1 - f)U_t + dE_t$
Employment: $E_{t+1} = (1 - d)E_t + fU_t$

- Consumption Euler equation: $u'(c) = \beta(1+r)u'(c^f)$ or generally $u'(c_t) = \beta(1+r)u'(c_{t+1})$
- Lifetime budget constraint $c + \frac{1}{1+r}c^f = a + y + \frac{1}{1+r}y^f$ or generally $\sum_{t=0}^T \left(\frac{1}{1+r}\right)^t c_t = a_0 + \sum_{t=0}^T \left(\frac{1}{1+r}\right)^t y_t$
- Optimality condition for investment:

$$p_k = \frac{1}{1+r} \left[(1 - \tau_K) MPK^f + p_k^f (1 - \delta) \right]$$

or

$$MPK^f = \frac{(1+r)p_k - (1 - \delta)p_k^f}{1 - \tau_K} = \text{user cost}$$

- Dynamics of debt-to-GDP ratio: $\Delta b_t = d_t + \frac{r-g}{1+g}b_{t-1}$ or $b_t = d_t + \frac{1+r}{1+g}b_{t-1}$